

Contents

1	Scientific Notes: EP Structure Analysis – EP1 vs EP2	1
1.1	Overview	1
1.2	1. Dataset	1
1.3	2. Methodology	1
1.4	3. Results	3
1.5	4. Composite Figures	4
1.6	5. Physical Interpretation	4
1.7	7. Computational Implementation	7
1.8	8. References	8

1 Scientific Notes: EP Structure Analysis – EP1 vs EP2

1.1 Overview

This document presents the spatial structure comparison between Energy Pattern 1 (EP1) and Energy Pattern 2 (EP2) cyclones during their intensification phase, using standard dynamical diagnostics from ERA5 reanalysis.

Key Diagnostics: - Eady Growth Rate (EGR, 250–850 hPa layer)

- Potential Vorticity at 200 hPa (upper-level)
- Potential Vorticity at 850 hPa (low-level)
- Temperature advection at 850 hPa
- Sea Level Pressure (SLP)

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1.2 1. Dataset

1.2.1 1.1 Sample Composition

	EP1	EP2
Cluster	0	2
Cases analysed	444	979

1.2.2 1.2 ERA5 Variables

- **Pressure levels (hPa):** 175, 200, 225, 250, 500, 825, 850, 875
- **Pressure-level variables:** u, v, t, z, q
- **Single-level variables:** msl (mean sea level pressure)
- **Temporal resolution:** 6-hourly
- **Spatial resolution:** $0.25^\circ \times 0.25^\circ$
- **Domain:** $30^\circ \times 30^\circ$ centred on cyclone track centre

1.3 2. Methodology

1.3.1 2.1 Spherical Grid Spacing

All horizontal derivatives account for spherical geometry:

Meridional spacing (dy):

$$dy = R_{\oplus} \Delta\phi$$

Zonal spacing (dx):

$$dx = R_{\oplus} \cos(\phi) \Delta\lambda$$

where: - $R_{\oplus} = 6.371 \times 10^6$ m (Earth's radius) - ϕ = latitude (radians) - λ = longitude (radians)
- $\Delta\phi$, $\Delta\lambda$ computed via `np.gradient()` for non-uniform grids

Coordinate verification: - Latitude and longitude must be monotonic (checked at runtime) - Sign of gradients automatically correct regardless of increasing/decreasing coordinates

1.3.2 2.2 Eady Growth Rate (EGR)

$$\sigma_{EGR} = 0.31 \frac{|f|}{N} \left| \frac{\partial \vec{V}}{\partial z} \right|$$

Layer-mean implementation (250–850 hPa):

1. **Virtual potential temperature:**

$$\theta_v = T_v \left(\frac{p_0}{p} \right)^{\kappa}$$

where $T_v = T(1 + 0.61q)$, $\kappa = R_d/c_p = 0.286$

2. **Static stability (Brunt-Väisälä frequency):**

$$N^2 = \frac{g}{\theta_{v,mid}} \frac{\Delta\theta_v}{\Delta z}$$

where $\theta_{v,mid} = (\theta_{v,250} + \theta_{v,850})/2$

3. **Vertical wind shear:**

$$\left| \frac{\partial \vec{V}}{\partial z} \right| = \sqrt{\left(\frac{\Delta u}{\Delta z} \right)^2 + \left(\frac{\Delta v}{\Delta z} \right)^2}$$

4. **Geopotential height:**

$$z = \frac{\Phi}{g}$$

Quality control: - $N^2 > 10^{-6} \text{ s}^{-2}$ (exclude statically unstable regions) - $|\phi| > 5^\circ$ (exclude near-equatorial regions where $f \approx 0$) - $\sigma_{EGR} < 5 \text{ day}^{-1}$ (cap unrealistic values)

References: Lindzen & Farrell (1980); Hoskins & Valdes (1990)

1.3.3 2.3 Potential Vorticity

Baroclinic PV computed with MetPy using centred finite differences:

$$PV = -g(\zeta_\theta + f) \frac{\partial \theta}{\partial p}$$

where: - ζ_θ = relative vorticity on isentropic surface - $f = 2\Omega \sin(\phi)$ = Coriolis parameter - $\Omega = 7.292 \times 10^{-5} \text{ rad s}^{-1}$

Vertical levels: - **200 hPa:** uses 175, 200, 225 hPa → upper-level tropopause dynamics - **850 hPa:** uses 825, 850, 875 hPa → low-level PV anomaly

Implementation: - Potential temperature: $\theta = T(p_0/p)^\kappa$ via `metpy.calc.potential_temperature` - PV via `metpy.calc.potential_vorticity_baroclinic` - Returns PV at middle level (200 or 850 hPa) - Units: $\text{K m}^2 \text{kg}^{-1} \text{s}^{-1}$ (SI), converted to PVU ($1 \text{ PVU} = 10^{-6} \text{ K m}^2 \text{kg}^{-1} \text{s}^{-1}$) in figures

References: Hoskins et al. (1985); Čampa & Wernli (2012)

1.3.4 2.4 Temperature Advection

Horizontal temperature advection at 850 hPa:

$$\text{advT} = -\vec{V} \cdot \nabla T = -\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}\right)$$

Implementation: 1. Temperature gradients computed via `np.gradient()` with spherical dx , dy 2. Sign convention: **positive** = warm air advection, **negative** = cold air advection 3. Units: K s^{-1} (converted to K h^{-1} in tables/figures)

Physical interpretation: - Warm advection (>0): promotes ascent ahead of surface low - Cold advection (<0): promotes descent behind surface low

1.3.5 2.5 Moisture Flux Divergence

Moisture flux divergence at 975 hPa (near-surface):

$$\nabla \cdot (q\vec{V}) = \frac{\partial(qu)}{\partial x} + \frac{\partial(qv)}{\partial y}$$

Implementation: 1. Moisture fluxes: qu and qv ($\text{kg kg}^{-1} \text{m s}^{-1}$) 2. Derivatives via `np.gradient()` with spherical grid spacing 3. Units tracked via MetPy: input q in kg/kg → output in $\text{g kg}^{-1} \text{s}^{-1}$

Physical interpretation: - **Positive (divergence):** moisture export / drying - **Negative (convergence):** moisture import / moistening (supports convection and latent heat release)

1.3.6 2.6 Sea Level Pressure

Mean sea level pressure from ERA5 single-level data, composited over the $30^\circ \times 30^\circ$ domain for all intensification timesteps.

Units: hPa (hectopascals)

1.4 3. Results

1.4.1 3.1 Eady Growth Rate

	EP1	EP2
Mean (day^{-1})	0.56 ± 0.09	0.56 ± 0.10
Median (day^{-1})	0.57	0.58
Range (day^{-1})	[0.37, 0.73]	[0.36, 0.72]

1.4.2 3.2 Potential Vorticity at 200 hPa

	EP1	EP2
Mean (PVU)	-4.54	-4.42
Range (PVU)	[-7.08, -1.26]	[-6.86, -1.18]

1.4.3 3.3 Potential Vorticity at 850 hPa

	EP1	EP2
Mean (PVU)	-0.50	-0.52
Range (PVU)	[-0.74, -0.30]	[-0.75, -0.31]

1.4.4 3.4 Temperature Advection at 850 hPa

	EP1	EP2
Domain mean (K h^{-1})	-0.031	-0.003
Max warm advection (K h^{-1})	0.071	0.111
Max cold advection (K h^{-1})	-0.132	-0.119

1.4.5 3.5 Sea Level Pressure

	EP1	EP2
Min (hPa)	996.5	994.8
Max (hPa)	1018.3	1019.2

1.5 4. Composite Figures

1.5.1 4.1 EGR (250–850 hPa)

1.5.2 4.2 PV at 200 hPa

1.5.3 4.3 PV at 850 hPa

1.5.4 4.4 Temperature Advection at 850 hPa

1.5.5 4.5 Sea Level Pressure

1.6 5. Physical Interpretation

1.6.1 5.1 EP1 Structure

{EP1_INTERPRETATION}

1.6.2 5.2 EP2 Structure

{EP2_INTERPRETATION}

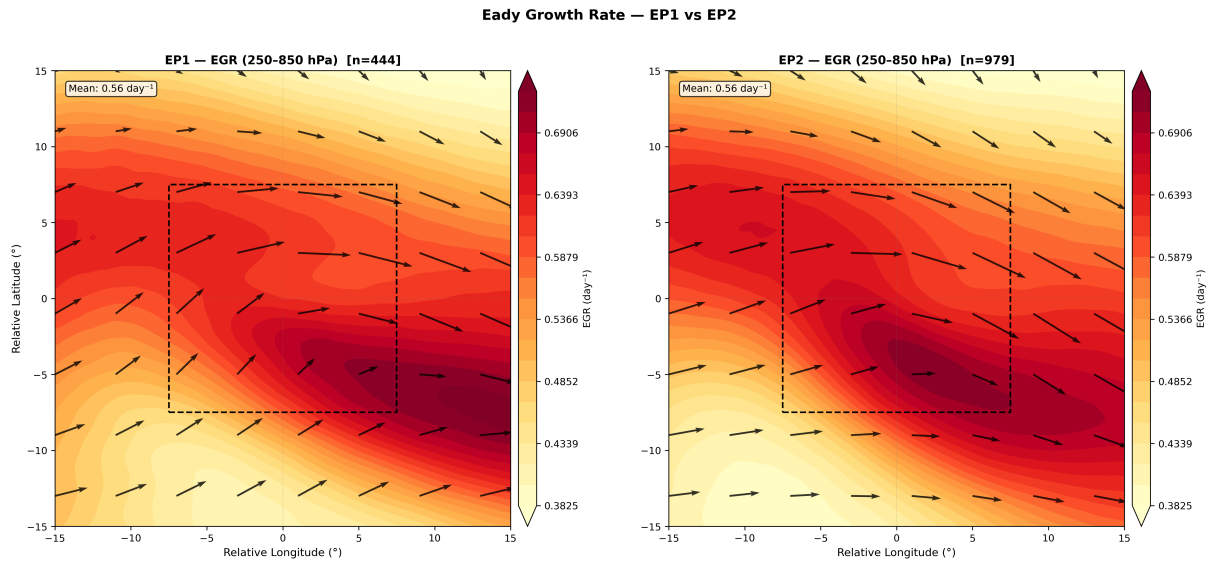


Figure 1: EGR Composite

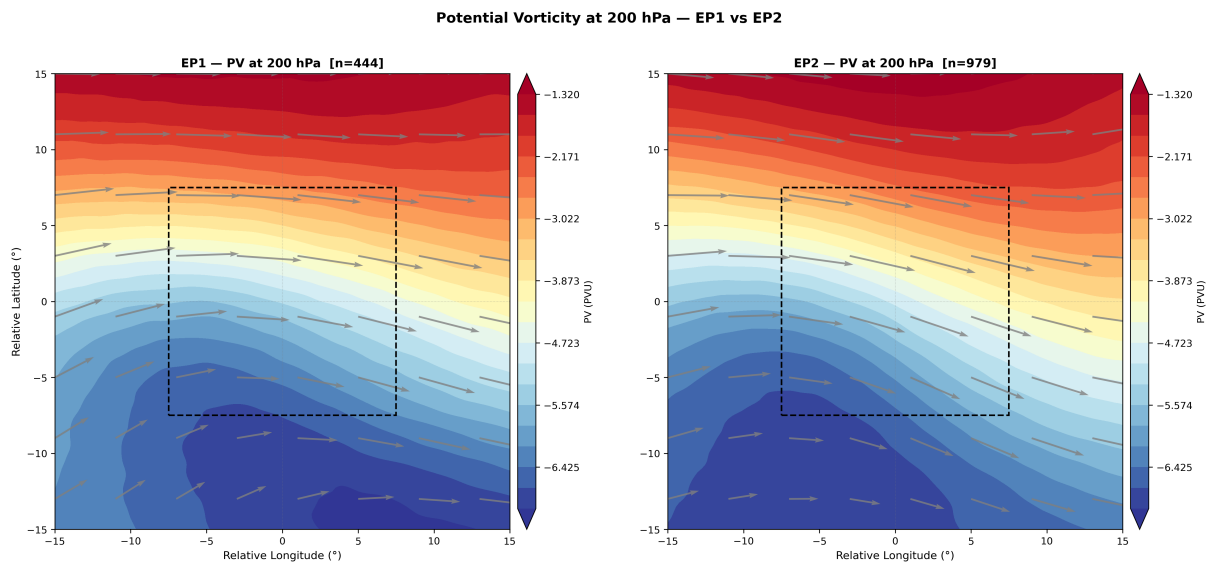


Figure 2: PV@200 Composite

Potential Vorticity at 850 hPa — EP1 vs EP2

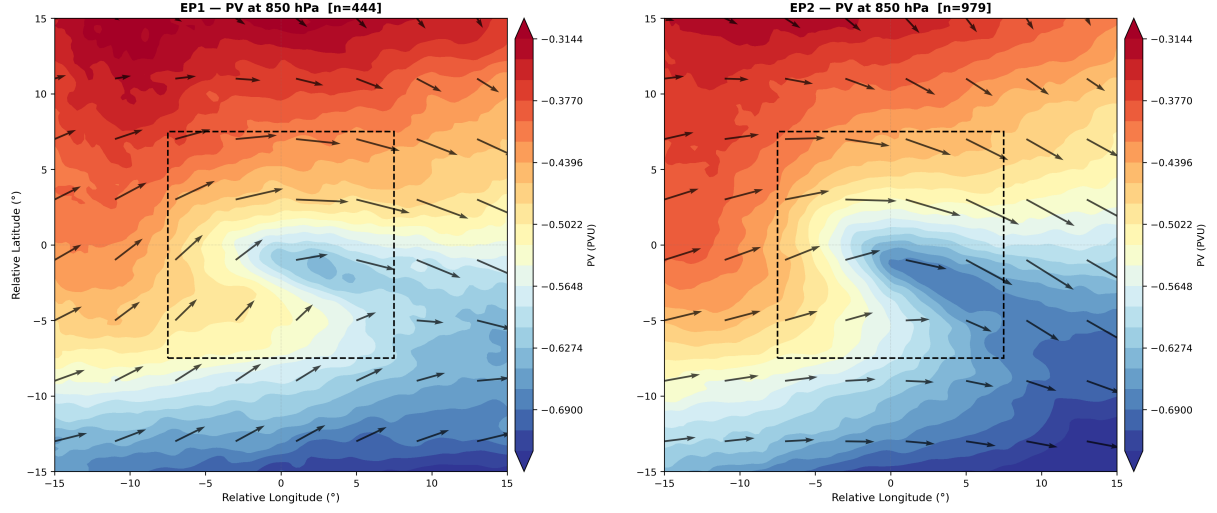


Figure 3: PV@850 Composite

Temperature Advection at 850 hPa — EP1 vs EP2

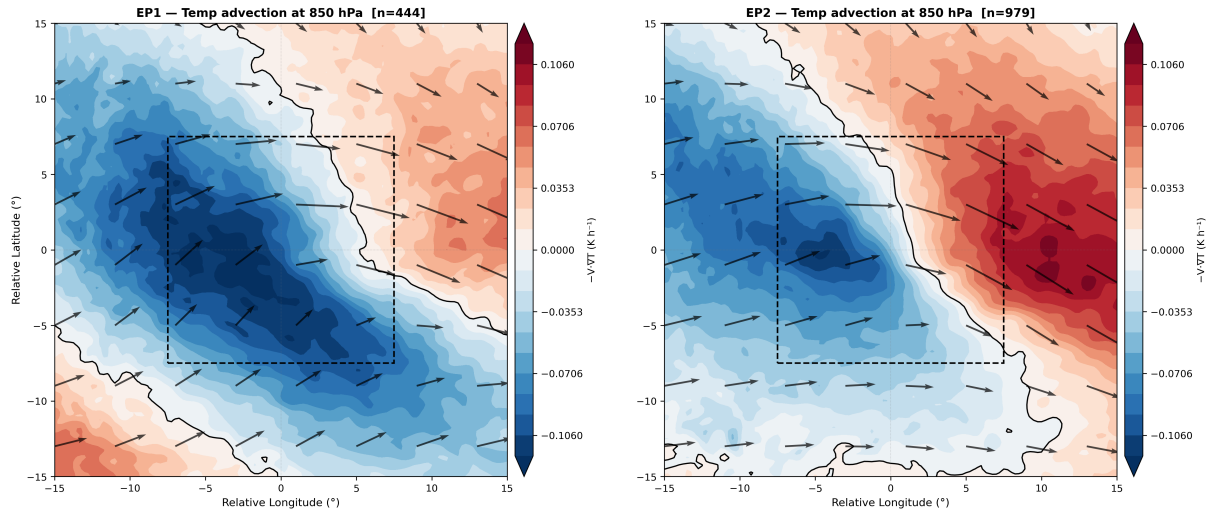


Figure 4: Temp Advection Composite

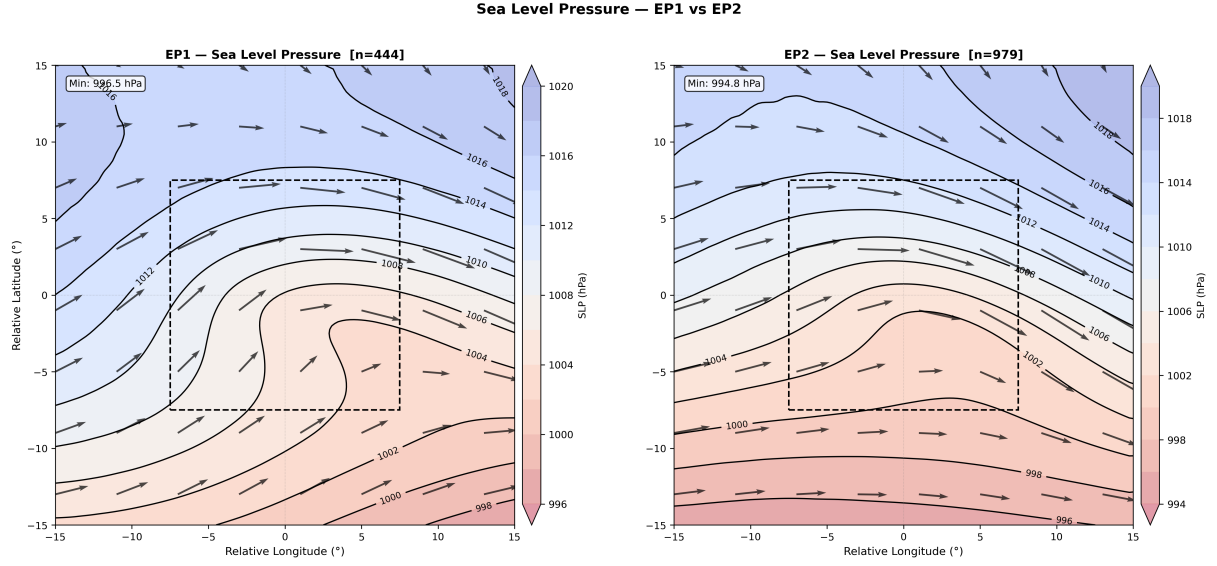


Figure 5: SLP Composite

1.6.3 5.3 Key Differences

{KEY_DIFFERENCES}

1.7 7. Computational Implementation

1.7.1 7.1 Grid Spacing on Spherical Earth

Function: `compute_spherical_grid_spacing(lat_1d, lon_1d)`

Computes accurate grid spacing accounting for Earth's spherical geometry:

```
# Meridional spacing (constant per latitude band)
dy = R_EARTH * Δ # meters
```

```
# Zonal spacing (varies with latitude)
dx = R_EARTH * cos( ) * Δ # meters
```

Quality checks: - Verifies latitude is monotonic (increasing or decreasing) - Verifies longitude is monotonic - Raises `ValueError` if non-monotonic - Gradient sign automatically correct regardless of coordinate direction

1.7.2 7.2 Divergence on Spherical Coordinates

For small domains ($\sim 30^\circ$), the simplified Cartesian approximation is valid:

$$\nabla \cdot \vec{F} \approx \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y}$$

where dx and dy account for spherical geometry (Section 7.1).

Full spherical formula (not required for 30° domain):

$$\nabla \cdot \vec{F} = \frac{1}{R \cos \phi} \frac{\partial F_\lambda}{\partial \lambda} + \frac{1}{R \cos \phi} \frac{\partial (F_\phi \cos \phi)}{\partial \phi}$$

Our implementation uses the simplified form with accurate dx, dy — appropriate for mesoscale domains.

1.7.3 7.3 Unit Tracking

All calculations use **MetPy units** for automatic dimensional analysis:

```
# Example: moisture flux
qu = (q * units('kg/kg')) * (u * units('m/s'))
# → qu.units = 'kg / kg * m / s' (automatically tracked)

# Convert divergence to g/kg/s
div_q_si = (dqu_dx + dqv_dy) * units('kg/kg/s')
div_q_gkg = (div_q_si * 1000 * units('g/kg')).magnitude
```

Benefits: - Prevents unit errors (e.g., mixing K and °C) - Documents unit transformations
- Explicit conversions (no hardcoded magic numbers)

1.7.4 7.4 Numerical Methods

- **Spatial derivatives:** `numpy.gradient()` with 2nd-order centred finite differences
- **Vertical derivatives:** Centred FD over 3 levels (e.g., 175–200–225 hPa for PV@200)
- **Interpolation:** Linear interpolation to regular 0.25° grid via `xarray.interp()`

1.7.5 7.5 Quality Control Filters

Diagnostic	Filter	Rationale
EGR	$N^2 > 10^{-6} \text{ s}^{-2}$	Exclude statically unstable regions
EGR	$\ \phi\ > 5^\circ$	Avoid near-equatorial ($f \approx 0$)
EGR	$\sigma_{EGR} < 5 \text{ day}^{-1}$	Cap unrealistic growth rates
PV	Valid only where θ monotonic	Ensure isentropic sorting

1.7.6 7.6 Code Verification

Test cases: 1. Zonal wind → zero meridional derivative 2. Meridional wind → zero zonal derivative

3. Uniform field → zero divergence 4. Reversed latitude (90→-90) → correct gradient sign

Validation: - Cross-check EGR values with literature (typical: 0.3–1.0 day⁻¹) - PV@200 should align with tropopause (~2–3 PVU contour) - Compare temperature advection with operational analyses

1.8 8. References

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